

Cacalol, a natural sesquiterpene, induces apoptosis in breast cancer cells by modulating Akt-SREBP-FAS signaling pathway

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Received: 8 March 2010 / Accepted: 17 July 2010 / Published online: 28 July 2010
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Abstract We previously isolated cacalol as a free radical-scavenging compound from *Cacalia delphiniifolia* which is a traditional Asian herbal plant and is believed to have medicinal effects on cancer. In this report, we demonstrated that cacalol has strong anti-proliferation effect on breast cancer cells and induces apoptosis by activating a pro-apoptotic pathway. We also found that a combination of cacalol and other chemotherapeutic drugs (Taxol and cyclophosphamide) synergistically induced apoptosis and partially overcame chemo-resistance. To further gain a mechanistic insight, we tested a potential inhibitory effect of cacalol on fatty acid synthase gene (FAS) in breast cancer cells, and found that cacalol significantly modulated the expression of the FAS gene, which resulted in apoptosis through activation of DAPK2 and caspase 3. We have also shown that cacalol significantly suppressed the Akt-sterol regulatory element-binding proteins (SREBP) signaling

pathway and concomitant transcriptional activation of FAS. In a xenograft model of nude mouse, when cacalol was administered intraperitoneally, tumor growth was significantly suppressed. Importantly, oral administration of cacalol before implanting tumors showed significant preventive effect on tumor growth in the same animal model. Furthermore, the treatment of mice with a combination of low dose of Taxol and cacalol significantly suppressed the tumor growth. Taken together, our results indicate that cacalol induces apoptosis in breast cancer cells and impairs mammary tumor growth in vivo by blocking the expression of the FAS gene through modulation of Akt-SREBP pathway, suggesting that cacalol has potential utility as a chemopreventive and chemotherapeutic agent for breast cancer.

Keywords Fatty acid synthase · Breast cancer · Herbal extract treatment · Apoptosis · Chemo-resistance

Abbreviations

FAS	Fatty acid synthase.
ROS	Reactive oxygen species.
SREBP	Sterol regulatory element-binding protein
CPA	Cyclophosphamide
HIF1	Hypoxia-inducible factor-1

Introduction

Fatty acid synthase (FAS), a complex multifunctional enzyme, is the sole enzyme capable of the reductive de novo synthesis of long-chain fatty acids from acetyl-CoA, malonyl-CoA, and nicotinamide adenine dinucleotide

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phosphate (NADPH) [1]. Although it is expressed at minimum or undetectable levels in most normal tissues, FAS is highly up-regulated in various types of human malignancies and is considered to be one of the most common molecular changes in cancer cells [2, 3]. In breast cancer, it has been reported that tumors marked by this prognostic molecule were nearly four times more likely to recur and metastasize than tumors not so marked [1, 4, 5]. In addition, since high levels of FAS have also been identified in pre-malignant lesions, FAS hyperactivation appears to be linked to neoplastic transformation. It has been demonstrated that the expression of cancer-associated FAS is induced by multiple oncogenic signalings including mitogen-activated protein kinase (MAPK ERK1/2) and PI3K-Akt pathways [6–13]. It has also been postulated that hyperactivation of FAS is in response to a microenvironment of tumors, thus conferring a selective growth advantage upon stresses such as hypoxia, low pH, and nutritional deprivation [14]. Recently, we have shown that the FAS gene is indeed up-regulated by hypoxia, which was also accompanied by reactive oxygen species (ROS) via activation of the Akt and hypoxia-inducible factor-1 (HIF1) followed by the induction of the sterol regulatory element-binding protein-1 (SREBP-1) gene in human breast cancer cells in vitro and in vivo [15]. These findings strongly suggest that the hyperactivation of FAS is actively contributing to the process of cell transformation rather than merely a consequence accompanied with the phenotypic changes of cancer cells. Importantly, treatment of tumor cells with pharmacologic inhibitors of FAS leads to cell cycle arrest followed by apoptosis. We have previously shown that specific blocking of the FAS expression by using siRNA in breast cancer cells caused an accumulation of malonyl-CoA, which led to the inhibition of carnitine palmitoyl transferase-1 (CPT-1) as well as increase in ceramide [16]. This was also followed by the induction of the proapoptotic genes, BNIP3, TRAIL, and DAP kinase 2 (DAPK2), which resulted in apoptosis of the tumor cells [16]. Therefore, high levels of FAS expression in cancer cells, the unambiguous association with tumor prognosis, and its unique enzymatic function of anti-apoptotic activity make FAS an attractive target for anti-tumor therapeutics as well as for chemoprevention, particularly with dietary supplements.

In the course of screening for free radical-scavenging compounds from edible plants, we isolated cacalol from *Cacalia delphiniifolia*, a traditional Asian herbal plant, which is believed to have medicinal effects on cancer [17]. Cacalol has also shown strong inhibitory activity against the lipid peroxidation induced by free radicals in a rat brain homogenate [17]. Many dietary phytochemicals have been shown to exhibit cancer-preventive and/or therapeutic

effects attributed to their antioxidant capacities. Therefore, in this report, we tested the hypothesis that cacalol, as an anti-oxidative component of *Cacalia delphiniifolia*, has an anti-tumor effect on breast cancer cells and examined its potential preventive and therapeutic utility. We found that cacalol significantly induced apoptosis in the tumor cells by inhibiting the FAS expression through modulation of Akt-SREBP pathway. This FAS-inhibitory activity of cacalol is mediated by the suppression of Akt-SREBP signaling pathway, which eventually led to apoptosis. Furthermore, we found that a combination of cacalol with other chemotherapeutic drugs, such as paclitaxel (Taxol) and cyclophosphamide (CPA), partially overcame chemoresistance of tumor cells. Importantly, cacalol, which was administered either intraperitoneally or orally, exhibited a significant inhibitory activity on the growth of mammary tumor without showing notable toxicity in a xenograft model.

Materials and methods

Cell culture and reagents

Human breast carcinoma cell lines, MCF7 and MDA-MB231, and non-tumorigenic mammalian epithelial cell lines, MCF10A and HBL-100 were obtained from American Type Culture Collection (Manassas, VA), and human mammary epithelial cell line (HMEC) was purchased from Lonza Inc. (Switzerland) and cultured in HEBM medium. paclitaxel (Taxol) resistant breast cancer cell line, MDA-MB231-TXL was isolated by our group. MDA-MB231-FASluc and MDA-MB231-luc are luciferase expressing cell lines that were derived from MDA-MB231 by stable transfection of the FAS promoter-driven and the SV40 promoter-driven luciferase gene, respectively. The cells were maintained in RPMI 1640 medium supplemented with 10% Fetal calf serum, streptomycin (100 µg/ml), penicillin (100 U/ml), and 250 nM dexamethasone (Sigma Chemical Co., St. Louis, MO) and grown at 37°C in a 5% CO₂ atmosphere. Hydrogen peroxide, *N*-acetyl-L-cysteine (NAC), CPA, paclitaxel (Taxol), 5-(tetradecyloxy)-2-furoic acid (TOFA) and fumonisins B1 were purchased from Sigma Chemical Co. LY294002 was obtained from Calbiochem (San Diego, CA). Cacalol was isolated from *Cacalia delphiniifolia* Sleb et Zucc as previously described [17]. In brief, *Cacalia delphiniifolia* Sleb et Zucc was freeze-dried and ground in a mixer. The resulting powder was extracted twice with a CH₂Cl₂-MeOH (1:1) solution. The extract was then subjected to silica gel chromatography using hexane-EtOAc (20:1) followed by HPLC chromatography.

Western blot

The cells were collected and resuspended in lysis buffer (50 mM Tris–HCl pH 7.4, 1% NP-40, 0.25% sodium deoxycholate, 150 mM NaCl, 1 mM EDTA). The lysates were boiled for 10 min, resolved by SDS-PAGE on an 8% polyacrylamide gel, and blotted onto nitrocellulose membrane. The membranes were treated with antibodies against FAS (0.2 µg/ml, Immuno-biological Laboratories Co., Japan), β -tubulin (1:1000, Upstate Biotechnology, Lake Placid, NY), SREBP-1 (1:200, Santa Cruz, Santa Cruz, CA), phospho-Akt (1:200, Ser473, Cell Signaling, Danvers, MA) and Akt (1:200, Cell Signaling). The membranes were then incubated with HRP-conjugated secondary antibodies and visualized by ECL Plus system (Amersham Life Sciences, Piscataway, NJ).

Quantitative real-time PCR

Total RNA was isolated from the cells and reverse-transcribed as previously described (16). The cDNA was then amplified with a pair of forward and reverse primers for the following genes: FAS (5'-CATCCAGATAGGCCTCATAGAC-3' and 5'-CTCCATGAAGTAGGAGTGGAAG-3'), DAPK2 (5'-CTTTGATCTCAAGCCAGAAAAC-3' and 5'-CTCGTAGTTCACAATTTCTGGAG-3'), CPT1 (5'-GCTTAGCTTCACGAGATCAGAC-3' and 5'-CCACAGATGGCTGGCATTTCAG-3'), PPAR γ (5'-GCTTCTACTGTGTGGGAACGAG-3' and 5'-CCCAATGAAGACAGCA GAAGAG-3') and β -actin (5'-TGAGACCTTCAACACCC CAGCCATG-3' and 5'-CGTAGATGGGCACAGTGTGG GTG-3'). PCR reactions were performed using DNA engine opticon2 system (MJ Research, MA) and the Dynamo SYBR Green qPCR Kit (Finnzyme Corp, Woburn, MA). The thermal cycling conditions comprised of an initial denaturation step at 95°C for 5 min followed by 35 cycles of PCR using the following profile: 94°C: 30 s, 58°C: 30 s, 72°C: 30 s.

In situ apoptosis assay

For detection of apoptosis in cultured cells, cells were grown in 96-well plates with or without treatment of cacalol for 48 h and fixed with 4% paraformaldehyde in PBS followed by permeabilization with 0.2% Triton-X 100/0.1% sodium citrate at 4°C. The cells were then washed extensively and TUNEL assay was performed using In Situ Cell Death Detection Kit/TMR Red (Roche Applied Science, Indianapolis, IN). The reaction was stopped after 1 h, and the number of apoptotic cells in each well was counted under a confocal microscope. For detection of apoptosis of paraffin-embedded tissue, deparaffinized and rehydrated sections were subjected to

permeabilization with 0.1% Triton-X 100/0.1% sodium citrate at 37°C in darkness for 4 h, after which they were washed twice with PBS. Endogenous peroxidase was reduced by treating sections in darkness with 3% H₂O₂ for 15 min, followed by two washes in PBS. Subsequent steps for TUNEL staining were carried out using In Situ Cell Death Detection Kit/POD (Roche Applied Science) according to the supplier's instructions. Following incubation of the sections with the TUNEL reaction mixture in a humidified chamber at 37°C for 1 h, TUNEL-POD solution was added at 37°C for 30 min. Sections were then treated with 3,3'-diaminobenzidine (DAB) substrate chromogen solution for 1–2 min (Envision-plus kit (Dako Corp. Carpinteria, CA) such that positive cells stained brown. Sections were counterstained with hematoxylin. To generate negative controls, terminal deoxynucleotidyl transferase (TdT) was replaced with sterile water, whereas for positive controls, sections were pre-treated with DNase I (10 µg/ml in distilled water) to generate DNA fragments.

Measurement of caspase 3 activity

The cells were grown in 24-well plates with or without treatment of cacalol for 48 h. Cells were collected and washed with PBS. Activity of caspase 3 was assayed using ApoAlert® Caspase-3 Colorimetric Assay Kit (Clontech Laboratories, Inc. Mountain View, CA) according to the manufacturer's protocol.

MTS assay

The cells were grown in 96-well plates with or without treatment of cacalol for 48 h. Cell survival was monitored using Cell Proliferation MTS Kit (Promega, Madison, WI) as recommended by the manufacturer. Absorbance values at 490 nm were recorded by an ELISA reader and the difference between these values was recorded as OD.

ROS assay

The cells were cultured with or without treatment of cacalol for 48 h, and fluorophore dichlorodihydrofluorescein diacetate (DCFDA, Sigma Chemical Co.) was added directly to the medium at a final concentration of 50 µM. The culture was further incubated at 37°C for 1 h and the cells were washed with PBS. The stained cells were visualized under a confocal microscope and photographed. The amount of staining was quantified by the MCID software.

Reporter assay

The luciferase reporter plasmid of the FAS promoter was described previously [15]. To generate deletion reporter

plasmids that contained –208, –74 or –53 FAS promoter region, the corresponding promoter regions were amplified with PCR and cloned into pGL3-basic. The luciferase reporter plasmids were then transfected to breast cancer cells, and luciferase activities were then measured by using Dual-Luciferase Reporter Assay System (Promega, Madison, MI) using Luminometer (Berthold Detection Systems, Huntsville, AL). For each transformation experiment, the Renilla expression plasmid pHRG-TK (Promega) was co-transfected as an internal control and promoter activities were normalized accordingly.

Chromatin immunoprecipitation

MDA-MB231 cells were cultured in T75 flask and fixed with 1% formaldehyde for 10 min at room temperature. To stop the reaction, 125 mM Glycine was added to the culture medium, and the cells were washed with PBS and harvested. Cells were then suspended in Cell lysis buffer (5 mM PIPE, 85 mM KCl, 0.5% Nonidet P-40) and homogenized with a type A Dounce homogenizer. The cell nuclei were collected and lysed with Nuclei lysis buffer (50 mM Tris–HCl, 10 mM EDTA, 1% SDS). The chromatin was sonicated on ice to an average length of 400 bp. The sample was then centrifuged at 4°C and the precipitates were resuspended in ChIP dilution buffer (16.7 mM Tris–HCl, 167 mM NaCl, 1.1% Triton X-100, 0.01% SDS). After pre-clearing the sample with Protein G agarose beads (DynaL Biotech, Invitrogen, Carlsbad, CA) followed by brief centrifugation, the supernatant was transferred to a new tube and anti-SREBP-1 (Santa Cruz) antibody was added. After overnight incubation at 4°C, Protein G agarose beads were added, and the sample was incubated for 3 h at 4°C. Beads were then washed with Washing buffer (100 mM Tris–HCl, 500 mM LiCl, 1% NP-40, 1% deoxycholic acid), and DNA–protein complexes were eluted with Elution buffer (100 mM NaHCO₃, 1% SDS). DNA–protein was de-crosslinked followed by phenol extraction, and the purified DNA was subjected to qPCR using both specific (F3:5′-GCGGCGCGCACGAGCATCACC-3′ and R1:5′-CGTCTCTCTGGCTCC CTCTAG-3′) and non-specific primers (F2:5′-CAGCCAGAGACACCTGTGGCC-3′ and R2:5′-CCTTTTCTGACCGCTTCGCGC-3′) for the SREBP binding/E-box sequence of the FAS promoter. The qPCR products were also visualized after electrophoresis on 8% acrylamide gel followed by staining with ethidium bromide.

Animal model

Breast cancer cell line, MDA-MB231luc, was suspended to 30 million/ml with PBS and equal volume of Matrigel (BD Biosciences, San Jose, California) was mixed with the

tissue. The cell suspension (0.1 ml) was injected into the mammary fat pad of 4-weeks-old female nude mice. The animals were treated with cacalol by either i.p injection or oral administration. For i.p injection, cacalol (30 mg/kg b.w.) or PBS was injected into the mice via i.p. starting 5 days after tumor inoculation and thereafter every 4 days for 3 weeks. To directly test the effect of cacalol on FAS expression in vivo, cacalol (10 mg/kg b.w.) or PBS was injected to mice via i.p. twice at day 9th and 11th after tumor inoculation. To examine the chemosensitizing effect of cacalol, a combination of cacalol (2.5 mg/kg b.w.) and Taxol (2 mg/kg b.w.) or each compound alone were injected into mice via i.p. every other day for four consecutive times. For oral administration of cacalol, the treatment was started 7 days before tumor cell inoculation, and cacalol (30 mg/kg b.w.) was given by a gavage every other day. The tumor size was measured for each mouse twice a week for 8 weeks using bioluminescent imaging (BLI) with an IVIS[®] Imaging System (Xenogen, Hopkinton, MA). For in vivo imaging, mice were given the substrate D-luciferin by i.p. injection at 150 mg/kg in PBS and anesthetized. Mice were then placed onto the warmed stage inside the light-tight camera box with continuous exposure to isoflurane. Imaging times ranged from 1 s to 15 s, and quantification of signals was controlled by the acquisition and analysis software Living Image[®] (Xenogen).

Immunohistochemistry

Five micrometer thick sections were cut out from the formaldehyde-fixed and paraffin-embedded tissue specimens and mounted on charged glass slides. The sections were baked at 60°C for 1 h, deparaffinized by two changes of xylene, and rehydrated in graded alcohol solutions. For antigen retrieval, the sections were heated in 10 mmol/l sodium citrate (pH 6.0) at 85°C for 30 min. The slides were treated with 3% H₂O₂ to block endogenous peroxidase activity and then incubated overnight at 4°C with anti-FAS rabbit polyclonal antibody (0.2 µg/ml; Immuno-biological Laboratories Co. Japan) or PBS as negative control. The sections were then incubated with horseradish peroxidase-conjugated anti-rabbit IgG for 30 min at room temperature, and 3,3′-diaminobenzidine(DAB) substrate chromogen solution [Envision-plus kit (Dako Corp.)] was applied. Finally, the sections were counterstained with hematoxylin.

Measurement of hepatic injury marker activities in plasma

At the end of the animal experiment, the mice were sacrificed. Plasma was prepared from blood by centrifugation at 1,750g for 15 min. Activities of alanine aminotransferase (ALT) and total bilirubin in plasma were measured

using commercial enzyme assay kits (Thermo Fisher Scientific, Waltham, MA).

Statistical analysis

For in vitro experiments, one-way and two-way ANOVA were used to calculate the P values. For all of the statistical tests, the significance was defined as $P < 0.05$. In all cases, Prism software was used.

Results

Cacalol treatment induces apoptosis in breast cancer cells

In order to examine a potential anti-proliferation effect of cacalol on breast cancer cells, we isolated cacalol from *Cacalia delphiniifolia* as previously described [23]. Cacalol is a strong antioxidative sesquiterpene ($C_{15}H_{18}O_2$) which chemical structure is shown in Fig. 1a [18]. We first examined the endogenous levels of ROS in MDA-MB231 cells that were cultured with or without cacalol, and found that cacalol treatment indeed significantly decreased the level of ROS (Fig. 1b, left panel). To further corroborate our results, we tested the effect of cacalol on the generation of ROS in the same cell line. As shown in Fig. 1b (right panel), the addition of 1 mM of H_2O_2 in the culture medium significantly augmented the generation of ROS ($P = 0.0065$); however, addition of cacalol significantly abrogated ROS level ($P < 0.0001$). We then treated MDA-MB231 and MCF7 with various doses of cacalol and monitored cell survival by MTT assay. As shown in Fig. 1c, cacalol significantly inhibited the growth of MDA-MB231 and MCF7 cells ($P = 0.0030, 0.0122, 0.0136, 0.0011, 0.0092$ for 5, 10, 17.5, 35, 70 μM in MDA-MB231, and 0.0079, 0.0025, 0.0006, 0.0001, 0.0001, 0.0001 for 0.1, 5, 10, 17.5, 35, 70 μM in MCF7, respectively). Furthermore, we examined the effect of cacalol on apoptosis by assessing DNA fragmentation in these cells. As shown in Fig. 1d, cacalol significantly induced apoptosis in MDA-MB231 and MCF7 cells, while only marginal number of cells underwent apoptosis in normal breast epithelial cells (HMEC) and non-tumorigenic breast epithelial cell lines (HBL100 and MCF10A) even at the high dose of cacalol treatment, indicating that cacalol induces apoptosis preferentially in malignant tumor cells while it is significantly less toxic to normal cells. Because caspase 3 is generally activated by its proteolytic cleavage at a late stage of apoptosis [19], we examined the status of caspase 3 activity in MDA-MB231 cells that were treated with or without cacalol. As shown in Fig. 1e, caspase 3 activity was significantly up-regulated by eight fold in MDA-MB231 cells

that were treated with cacalol ($P = 0.0009$). These results indicate that cacalol exerts its anti-proliferation activity in tumor cell-specific manner by activating the caspase pathway followed by induction of apoptosis.

Cacalol synergistically induces apoptosis with Taxol and CPA in breast cancer cells

Resistance to chemotherapy is a general concern for the treatment of malignant diseases. We therefore examined whether cacalol affects the responsiveness of chemo-resistant cells to chemo-therapeutic drugs. MDA-MB231 and MDA-MB231-TXL cells were cultured and treated with low (10 μM) and high (40 μM) doses of Taxol followed by treatment with or without cacalol (17.5 μM). As shown in Fig. 2a, Taxol treatment induced apoptosis in a dose-dependent manner in MDA-MB231 cells (left panel) but not in MDA-MB231-TXL cells (middle panel); however, addition of cacalol induced significantly higher level of apoptosis in both cell lines. Notably, the apoptotic effect of the combination of Taxol and cacalol significantly exceeded the sum of either agent administered alone, indicating a significant synergistic effect of these two agents ($P < 0.0001$, as examined by two-way factorial ANOVA). These results suggest the potential utility of cacalol at low concentrations with a combination of other drugs as a rational strategy for the treatment of chemo-resistant tumors. The down-regulation of FAS expression by cacalol in both cell lines was confirmed by western blot as shown in Fig. 2a (right panel). Furthermore, we examined the tumoricidal effect of cacalol in combination with CPA which is commonly used for breast cancer therapy. MDA-MB231 cells were cultured and treated with various amounts of CPA in the presence or absence of a low concentration of cacalol (3.5 μM) for 48 h. Western blot analysis was then performed to examine the expression of FAS. As shown in Fig. 2b (left panel), the combination of these two compounds at the low concentration strongly suppressed FAS, while the same concentration of each compound alone did not show significant inhibitory effect on FAS. Furthermore, we found that the combination of CPA with cacalol-induced apoptosis in MDA-MB231 cells in a synergistic manner compared to the treatment with CPA or cacalol alone (Fig. 2b, right panel, $P < 0.0001$, as examined by two-way factorial ANOVA). Therefore, these results strongly suggest a potential therapeutic utility of cacalol in combination with conventional chemotherapeutic agents.

Cacalol inhibits FAS expression, followed by induction of apoptosis

Since cacalol preferentially induced apoptosis in breast cancer cells and it showed significantly less toxicity to

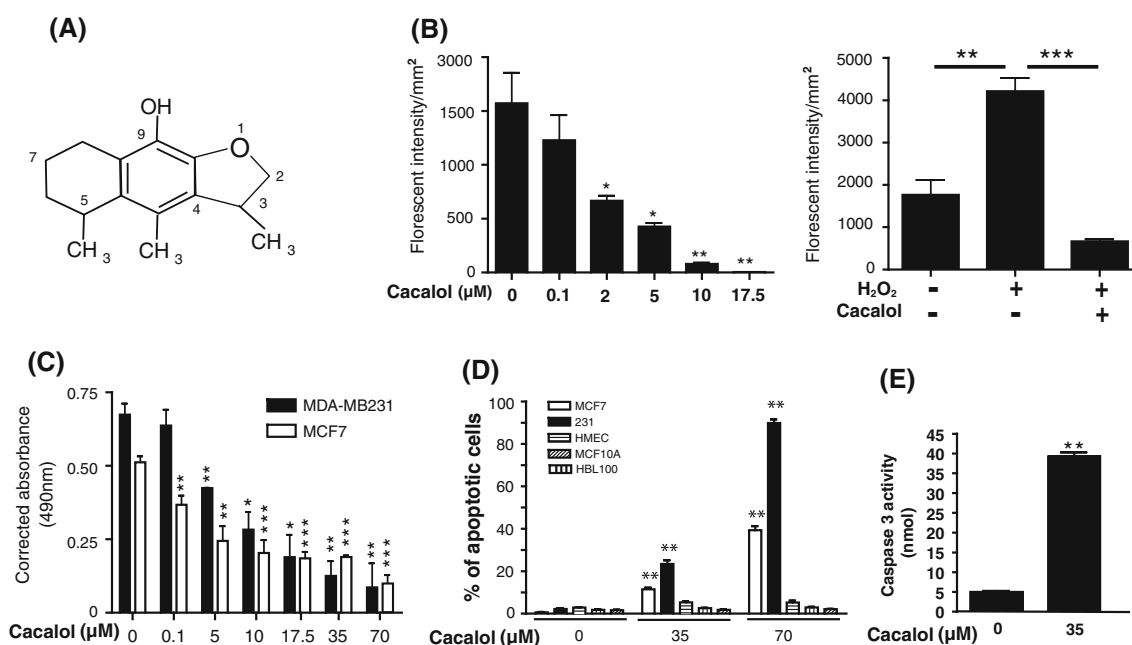


Fig. 1 Cacalol induces apoptosis in breast cancer cells. **a** Chemical structure of cacalol. **b** MDA-MB231 cells were cultured with or without cacalol for 48 h, and cells were then treated with the DCFDA dye for assaying the amount of ROS. The stained cells were visualized under a confocal microscope and photographed. The amount of staining was quantified by using the MCID software (left panel). MDA-MB231 cells were treated with or without H₂O₂ (1 mM) and/or cacalol (10 μM) for 24 h. ROS level was examined as above (right panel). Values are means ± SD of triplicate measurements. **c** MCF7 and MDA-MB231 cells were treated with or without various amounts of cacalol for 48 h at 37°C. Cell survival was

monitored by MTT assay. **d** MCF7, MDA-MB231, MCF10A, HBL100, and human mammary epithelial cells (HMEC) were cultured and treated with or without cacalol at different concentrations (35 and 70 μM) for 48 h at 37°C, and then apoptosis assay was performed by using In situ cell death TMR red kit. Apoptotic cells were visualized and counted under a confocal microscope. Values are means ± SD of triplicate measurements. **e** MDA-MB231 cells were treated with or without cacalol (35 μM) for 48 h at 37°C and then harvested. Cell lysates were tested for caspase 3 activity by ApoAlert Kit

normal cells, we reasoned that there was a specific target for cacalol which may be activated in malignant cells. FAS is considered as one of the most common molecular changes in cancers as it is expressed at minimum or undetectable levels in most normal tissues, while it is highly up-regulated in various types of human cancers. Therefore, we hypothesized that cacalol affects FAS expression followed by induction of apoptosis in breast cancer cells. We previously reported that FAS expression was partly controlled by hypoxia through the generation of reactive oxygen species (ROS) which is often produced at a high level under hypoxic microenvironment of a growing tumor [15]. Therefore, we first tested whether cacalol antagonizes ROS-induced FAS expression in breast cancer cells. As shown in Fig. 3a, FAS expression was significantly up-regulated in response to H₂O₂ ($P = 0.003$), while cacalol treatment restored FAS expression to the basal level even in the presence of H₂O₂ ($P = 0.0042$). To test whether cacalol has a direct inhibitory effect on FAS expression instead of merely working as a free radical scavenger, RNA was isolated from MDA-MB231 cells that were cultured in the presence or absence of cacalol, and they were subjected to quantitative reverse transcription-

PCR (qRT-PCR). As shown in Fig. 3b (left panel), cacalol significantly inhibited the expression of FAS gene ($P = 0.0353$) while it did not affect two other genes, CPT-1 and PPAR γ (peroxisome proliferator-activated receptor γ), that are involved in lipogenesis, suggesting the specificity of the cacalol's effect on the FAS expression at transcriptional level. Furthermore, we examined the effect of cacalol on the promoter activity of the FAS gene in MDA-MB231-FASluc which contains a stably transfected FAS promoter-driven luciferase gene. As shown in Fig. 3b (right panel), cacalol treatment induced significant reduction of FAS promoter activity in breast cancer cells ($P = 0.0083$). We also examined the effect of cacalol on the expression of FAS at the protein level by culturing two breast cancer cell lines, MDA-MB231 and MCF7, in the presence or absence of cacalol at various doses for different time periods followed by western blot analysis. As shown in Fig. 3c, cacalol inhibited the expression of FAS in dose- and time-dependent manners in both cell lines. Previously, we demonstrated that apoptosis caused by the inhibition of FAS is due to up-regulation of a series of pro-apoptotic genes (DAPK2, BNIP3, and TRAIL) which was mediated by the accumulation of malonyl-CoA and ceramide [16].

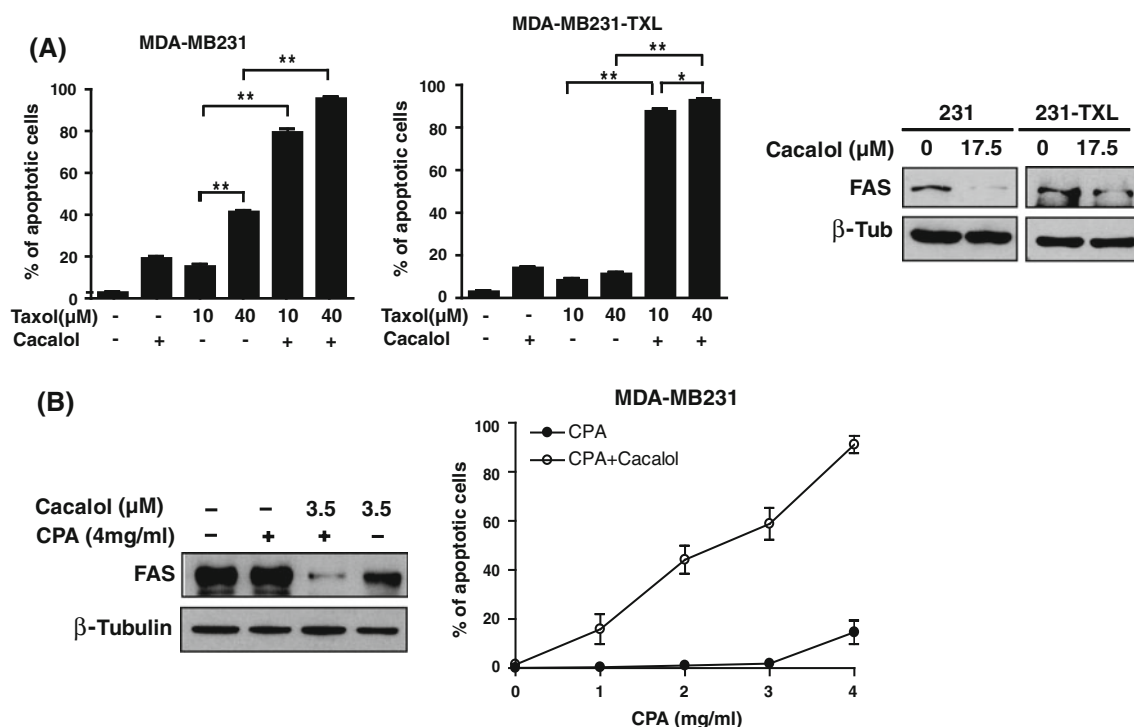


Fig. 2 Cacalol synergistically induces apoptosis with Taxol and CPA in breast cancer cells. **a** MDA-MB231 and MDA-MB231-TXL cells were cultured and treated with Taxol (10 μM and 40 μM) in the presence or absence of cacalol (17.5 μM) for 48 h at 37°C. Cells were then subjected to the apoptosis assay by using in situ cell death TMR red kit. FAS expression in both cell lines treated with cacalol were

examined by western blot. **b** MDA-MB231 cells were cultured and treated with various amounts of CPA in the presence (*open circle*) or absence (*closed circle*) of cacalol (3.5 μM) for 48 h, followed by western blot analysis for the expression of FAS. Cells were then subjected to apoptosis assay. Apoptotic cells were visualized and counted under a confocal microscope

To further determine that cacalol-induced apoptosis is dependent on the inhibition of FAS, TOFA, a specific inhibitor of acetyl-CoA carboxylase that catalyzes synthesis of malonyl-CoA from acetyl-CoA, and fumonisins B1 which is a ceramide synthase inhibitor, were used in combination with cacalol to treat MDA-MB231 cells. As shown in Fig. 3d, both TOFA and fumonisins B1 significantly blocked the ability of cacalol to induce apoptosis. Moreover, we found that treatment of MDA-MB231 cells with 35 μM cacalol for 48 h indeed significantly enhanced the expression of the DAPK2 gene (Fig. 3e, $P = 0.0014$). Collectively, our results indicate that cacalol is capable of inhibiting the messenger RNA and protein expression of FAS in breast cancer cells, which significantly induces apoptosis in these cells.

Cacalol inhibits FAS expression through interfering with Akt-SREBP-FAS signaling pathway

Next, because it was previously reported that hypoxia induces FAS expression through Akt-SREBP pathway, we examined if cacalol is capable of interfering with the Akt-SREBP signaling [15]. As shown in Fig. 4a, we found that both cacalol and LY294002, a PI3K inhibitor, significantly

blocked FAS expression ($P = 0.0229$). We then examined the status of Akt and SREBP in the presence or absence of cacalol in MDA-MB231 cells. As shown in Fig. 4b, our results of western blot analysis indicate that the expression of FAS and SREBP-1 were indeed down-regulated in response to cacalol. The amount of phospho-Akt (Ser473) was also strongly decreased in the same set of samples treated with cacalol, while the amount of total Akt was not affected. SREBP is known as the major transcriptional regulator of the FAS gene during the lipogenesis, and we previously showed that hypoxia induced the FAS gene by activating Akt followed by induction of SREBP-1 binding to the FAS promoter [15]. To identify the cacalol-responding sequence on the FAS gene promoter, we generated a series of luciferase reporter plasmids containing up to -208, -74 and -53 base of the FAS promoter, and luciferase reporter activities were measured in the presence or absence of cacalol. As shown in Fig. 4c (upper panel), the deletion of the SREBP/E-box elements (SRE) sequence at -150 base of the FAS promoter region did not affect the responsiveness of the FAS promoter to cacalol treatment ($P = 0.0257$ and 0.0052 , respectively). However, further deletion of the two SRE sequences at -65 base of the FAS promoter region significantly inhibited the reduction of the

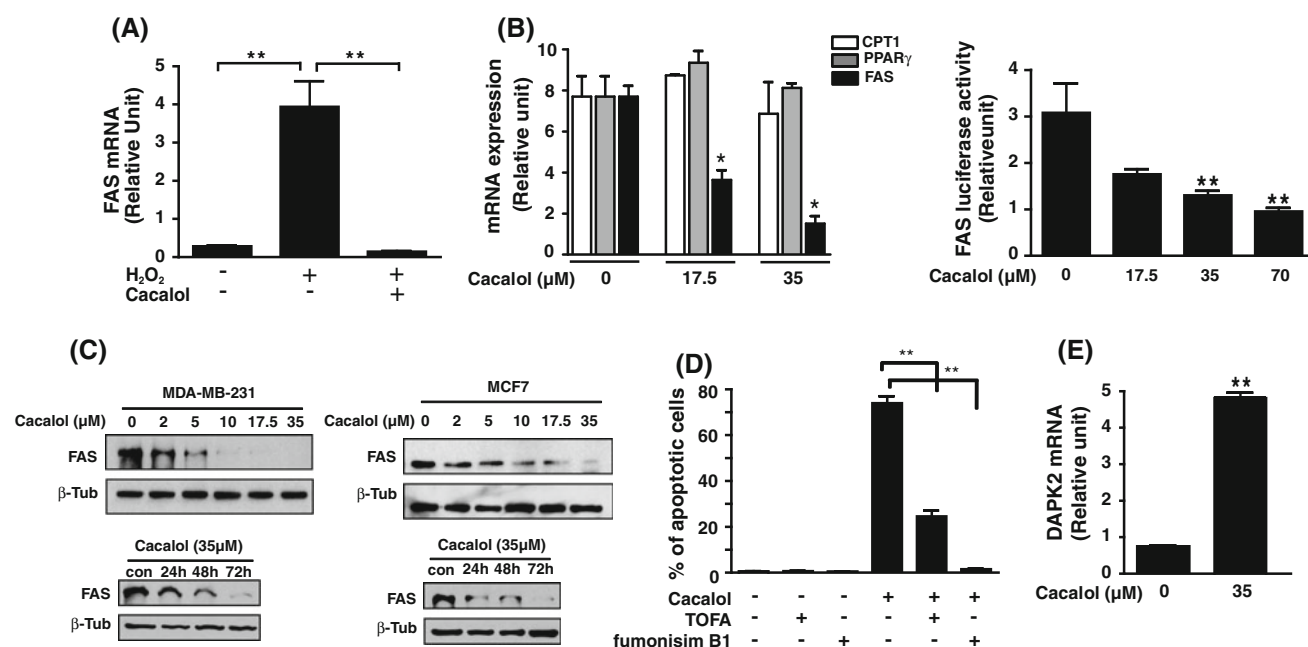


Fig. 3 Cacalol inhibits FAS expression, followed by induction of apoptosis. **a** MDA-MB231 cells were treated with or without H₂O₂ (1 mM) and/or cacalol (35 μ M) for 24 h. FAS expression was measured by qRT-PCR (right panel). Values are means $\pm$ SD of triplicate measurements. **b** MDA-MB231 cells were treated with or without cacalol for 48 h at 37°C. Cells were then harvested and subjected to qRT-PCR using primers for the FAS, CPT-1, PPAR γ and β -actin genes (left panel). MDA-MB231-FASluc cells were treated with or without cacalol for 48 h at 37°C. Cells were then harvested and subjected to luciferase assay (right panel). Values are means $\pm$ SD of triplicate measurements. **c** Human breast cancer cell lines MCF7 and MDA-MB231 were treated with cacalol at various

doses (for 48 h) and for different time periods (at 35 μ M), and cell lysates were subjected to western blot using anti-FAS antibody. Anti- β -tubulin was used as control. **d** MDA-MB231 was cultured and treated with or without cacalol (70 μ M) in combination with TOFA (30 μ M) and fumonisins B1 (50 μ M) for 48 h at 37°C, and then the apoptosis assay was performed by using In situ cell death TMR red kit. Apoptotic cells were visualized and counted under a confocal microscope. Values are means $\pm$ SD of triplicate measurements. **e** MDA-MB231 was also treated with or without cacalol (35 μ M) for 48 h at 37°C. Cells were harvested and subjected to qRT-PCR using primers for the DAPK2 and β -actin genes. Values are means $\pm$ SD of triplicate measurements

FAS promoter activity in response to the cacalol treatment, indicating that the SRE sequence at -65 in the FAS promoter region is essential for the inhibitory effect of cacalol. To further determine whether cacalol suppresses the SREBP binding to the FAS promoter, we performed ChIP assay by precipitating SREBP–chromatin complex using anti-SREBP-1 antibody followed by qPCR with the primers specific to the region of the SREBP-binding site. Our result clearly indicates that cacalol treatment significantly blocks the binding of SREBP-1 to the FAS promoter (Fig. 4c, lower panel), suggesting that cacalol suppresses the FAS expression by modulating the Akt/SREBP-1 pathway.

Cacalol suppresses the mammary tumor growth in vivo

The strong inhibitory effect of cacalol on tumor cells in our in vitro experiments suggests a potential clinical utility of this compound. Therefore, we examined the efficacy of cacalol on breast cancer growth in a xenograft model by transplanting human breast cancer cell line, MDA-MB231-luc, into nude mice. After 5 days of injection of the tumor

cells, we began to inject cacalol (30 mg/kg b.w.) into one group of mice and EtOH/PBS in the other group as control via i.p. followed by injection of cacalol or EtOH/PBS every 4 days (day 9, 13, 17, 21, 25 as indicated by arrows in Fig. 5a right panel) for 3 weeks. We periodically monitored the growth of tumor using bioluminescent imaging for 8 weeks. As shown in Fig. 5a right panel, the tumors grew progressively from week 3 and were significantly larger in control mice than that in cacalol-treated mice as reflected by the signal strength of the luciferase activity ($P = 0.0365, 0.0284, 0.0121, 0.0005$ at day 25, 35, 45 and 50, respectively). The representative images of the tumor growth in both groups at day 50 were shown in Fig. 5a (left panel). Notably, mice in both groups had comparable body weight during the course of the experiment (data not shown). At the endpoint, mice were sacrificed and examined for the expression of FAS in the tumor by immuno-histochemical analysis. As expected, we found that FAS was highly expressed in tumors from control mice, whereas it was substantially decreased in tumors from cacalol-treated mice (Fig. 5b, upper panels). Moreover, tumors of cacalol-treated mice showed substantially higher level of

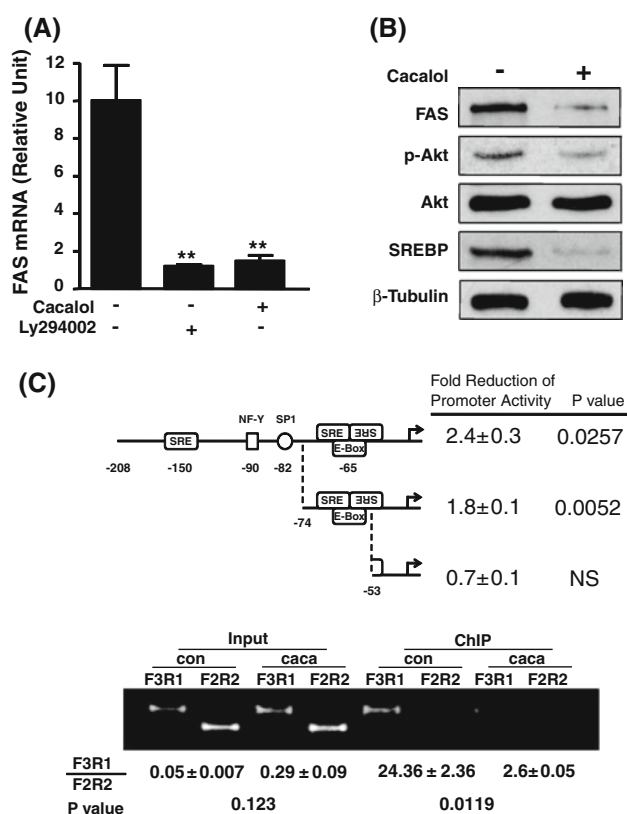


Fig. 4 Cacalol inhibits Akt-SREBP-FAS signaling pathway. **a** MDA-MB231 cells were treated with cacalol (35 μ M) and LY294002 (20 μ M) for 48 h, followed by qRT-PCR analysis for the FAS gene expression. Values are means $\pm$ SD of triplicate measurements. **b** MDA-MB231 cells were cultured and treated with or without cacalol (35 μ M) for 48 h at 37°C. Cell lysates were subjected to western blot using antibodies for FAS, SREBP-1, phospho-Akt (p-Akt), total Akt (Akt) and β -tubulin. **c** Luciferase reporter constructs with various lengths of the FAS promoter were transfected to tumor cells that were then cultured with or without cacalol (35 μ M) for 48 h at 37°C. Cells were then collected and assayed for luciferase activities (upper panel). For ChIP assay, MDA-MB231 cells were cultured with or without cacalol (35 μ M) for 48 h. The cells were lysed and the lysate was pulled down with anti-SREBP-1 antibody. The DNA was then subjected to qPCR using non-specific (F2, R2) or SREBP-binding site-specific primers (F3, R1). The ratio of the DNA was calculated based on cyclic threshold value for each reaction (right panel). Values are means $\pm$ SD of triplicate measurements

apoptosis than that of un-treated mice (Fig. 5b, lower panels). These results strongly indicate that the inhibition of FAS induced cell apoptosis in cacalol-treated mice and eventually led to tumor growth suppression, which is consistent with our in vitro data. To examine a potential toxicity of cacalol, we collected the plasma at the end of the experiment, and activities of alanine aminotransferase (ALT) and total bilirubin in plasma were measured using commercial assay kits. As shown in Fig. 5c, the values for both ALT and total bilirubin were within normal ranges for cacalol-treated mice compared to the control mice and normal mice without injection of tumor cells, indicating

that cacalol does not have notable toxicity to mice. To further support our hypothesis that FAS is a down-stream target for cacalol-induced tumor suppression in vivo, we treated mice with lower dose of cacalol (10 mg/kg b.w.) or PBS and sacrificed them 12 days after tumor cell transplantation. As shown in Fig. 5d, we found that the FAS expression in tumors of cacalol-treated mice was significantly reduced at both mRNA and protein levels compared with those of PBS-treated mice ($P = 0.0054$), although tumor size were similar in both groups ($106.9 \pm 21.85 \text{ mm}^3$ for cacalol-treated group and $110.6 \pm 42.54 \text{ mm}^3$ for PBS-treated group). Furthermore, the strong chemosensitizing activity of cacalol shown in our in vitro experiments prompted us to examine the effect of a combination of Taxol and cacalol in our orthotopic xenograft animal model. One week after the transplantation of MDA-MB231-luc tumor cells, mice were treated with Taxol (2 mg/kg b.w.) either alone or combined with a lower dose of cacalol (2.5 mg/kg b.w.) every other day for four times. We found that at day 33, mice that received the combination therapy showed significantly stronger reduction of tumor growth and FAS expression compared with the mice treated with either compound alone (Fig. 5e, $P = 0.0233$ for Cacalol vs Cacalol + TXL, and 0.0138 for TXL vs. Cacalol + TXL). In addition, to examine a potential preventive effect of cacalol by oral administration, we treated mice every other day with cacalol (30 mg/kg b.w.) using a gavage cannula. One week after cacalol administration, MDA-MB-231-luc cells were orthotopically injected into nude mice and the oral administration of cacalol was continued. As shown in Fig. 5f (left panel), we found that orally given cacalol significantly delayed the onset of tumor in nude mice compared with control mice ($P = 0.0203$). Of note, no significant weight loss was observed in the cacalol-treated groups (Fig. 5f, right panel). Taken together, our results strongly suggest the utility of cacalol for both prevention and therapeutic application for breast cancer.

Discussion

Cacalol has been isolated from various medicinal plants including *Cacalia delphiniifolia*, which is a popular edible wild plant in Asian countries [20, 21]. Cacalol was also identified as major anti-hyperglycemic and anti-inflammatory sesquiterpene in the aqueous extract from the roots of *Psacalum decompositum*, which has been used as a traditional medicine in Mexico and the Southwestern United States for the treatment of rheumatism, fever, skin infections and ulcers, diabetes and tumors [22, 23]. In this report, we have shown that cacalol has antitumor activity as it significantly inhibited breast cancer cell growth and induced apoptosis, and more importantly, significantly

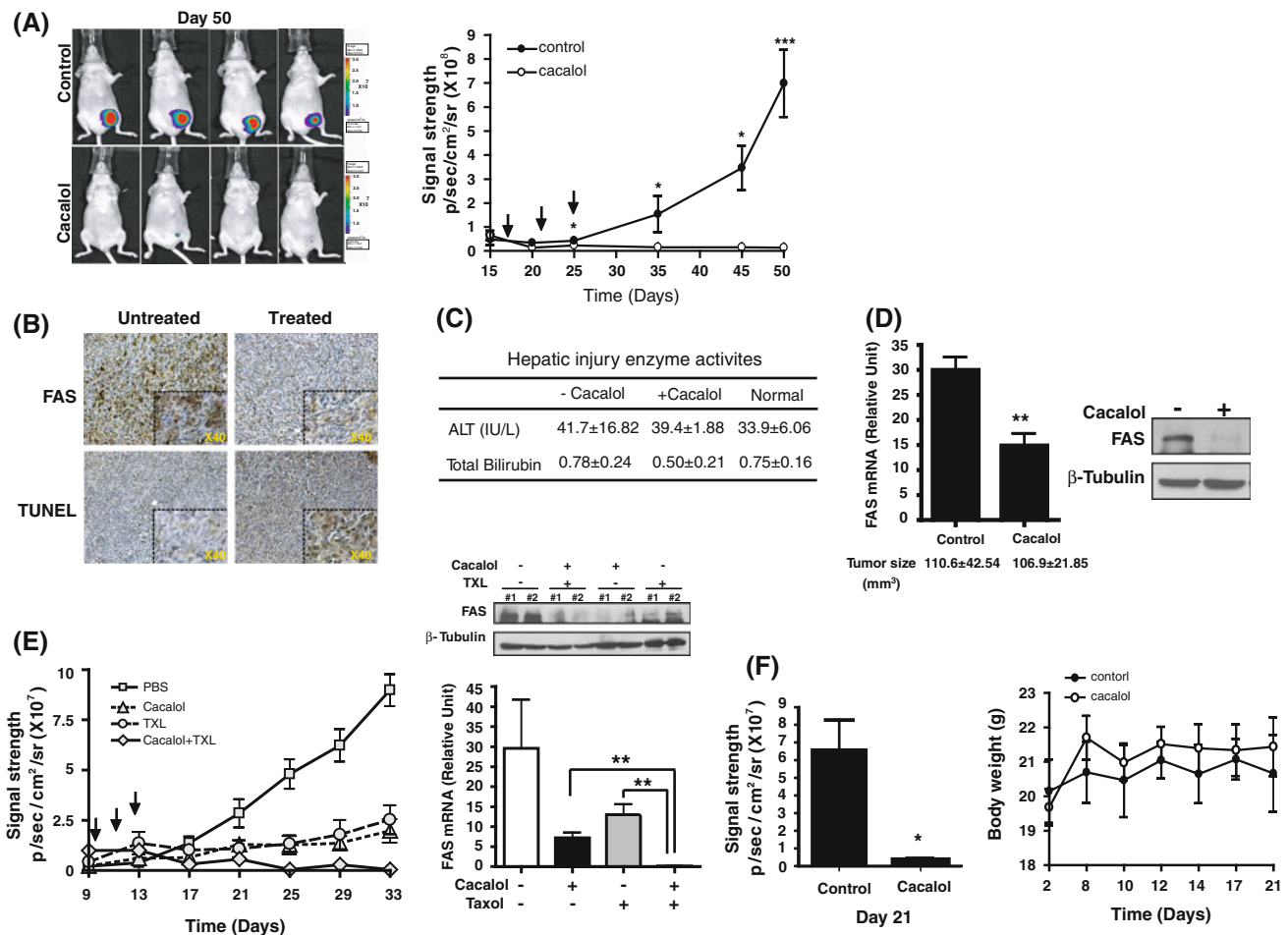


Fig. 5 Inhibitory effect of cacalol on mammary tumor growth in mice. **a** Human breast cancer cell, MDA-MB231-luc, was injected orthotopically into 4-weeks-old female nude mice ($n = 5$). Cacalol (30 mg/kg b.w.) was injected intraperitoneally to the mice every 4 days for 3 weeks. The tumor size was measured for each mouse twice a week for 8 weeks using bioluminescent imaging with an IVIS Imaging System (Xenogen). Imaging and quantification of signals were controlled by the acquisition and analysis software Living Image. **b** At the end of the experiment, the mice were sacrificed, tumors were excised and stained with antibody for FAS (upper panels). The degree of apoptosis in the same area of tumor sections was examined by In situ cell death POD kit. Apoptotic cells were visualized by immunohistochemical staining (lower panel). **c** The activities of hepatic injury markers, alanine aminotransferase (ALT) and total bilirubin in plasma were measured using commercial assay kits. **d** Mice were treated with cacalol (10 mg/kg b.w.) or PBS

(control) at day 9 and 11 after tumor inoculation. Mice were sacrificed the next day and the FAS expressions of tumors was measured by qRT-PCR and western blot. **e** One week after orthotopic injection of MDA-MB231-luc, nude mice were treated with cacalol (2.5 mg/kg b.w.) and Taxol (2 mg/kg b.w.) either alone or in combination. Tumor growth was monitored by measuring luciferase activity. FAS expression at the end of experiment was measured by qRT-PCR and western blot. **f** Four-weeks old female nude mice were treated with cacalol (30 mg/kg b.w.) by oral administration every 2 days for 1 week ($n = 5$). MDA-MB231-luc cells were then orthotopically inoculated to the animals and the oral cacalol treatment was continued. The tumor growth was monitored periodically for 3 weeks using bioluminescent imaging with an IVIS Imaging System. Weight of mice with or without the treatment of cacalol was periodically measured

suppressed mammary tumor growth in a xenograft mouse model without showing apparent toxicity through either oral or i.p. administration. Interestingly, we also found that the crude extract of *C. delphiniifolia* significantly blocked FAS expression and induced apoptosis in breast cancer cells but not in normal mammary epithelial cells. Considering the non-toxicity of this plant as it has been consumed in routine diet, it may have the potential utility as a dietary supplement for chemopreventive medicine.

As first demonstrated more than half a century ago, many human cancers display alterations in lipid metabolism and the high rates of *de novo* fatty acid biosynthesis despite adequate nutritional supply [24, 25]. Recent evidence strongly indicates that the hyperactivation of FAS, an anabolic energy storage pathway of minor importance in normal cells, is actively contributing to the process of cell transformation rather than merely a consequence accompanied with the phenotypic changes of tumor cells [1–3, 24, 26].

Furthermore, inhibition of FAS by pharmacologic agents or by genetic means has been shown to reduce proliferation rates and to induce apoptosis in tumor cells [27–29]. FAS is therefore considered as an ideal target for anticancer therapy. Here we demonstrated that cacalol acts as a potent inhibitor of FAS through modulation of Akt-SREBP pathway in human breast cancer cells as it significantly suppresses the expression of the gene in a wide range of breast cancer cell lines even in chemo-resistant cells. Importantly, we found that the inhibition of FAS by cacalol-induced apoptosis in breast cancer cells but not in normal epithelial cells.

The expression of FAS was reported to be responsive to the microenvironment of solid tumors containing regions of poor oxygenation and high acidity due to a lack of adequate angiogenesis and/or nutritional supply, and therefore, FAS functions as a “metabolic oncogene” conferring a selective growth advantage upon stresses such as hypoxia, low pH, and nutritional deprivation [14]. Our previous results indeed showed that FAS expression was significantly enhanced by hypoxia and H₂O₂ through activation of HIF1 and Akt (Ser⁴⁷³), and consequently up-regulated SREBP-1, a major transcriptional factor involved in the regulation of the FAS gene [15]. In the present report, we have shown that cacalol strongly suppressed the activation of Akt (Ser⁴⁷³) and SREBP-1, followed by significantly reduced the amount of SREBP-1 binding to the FAS promoter. The deletion of SRE sequences on the promoter significantly abolished the FAS-inhibitory activity of cacalol. Therefore, cacalol suppresses the expression of FAS by overall inhibition of SREBP-1 activity through the blockade of PI3K/Akt signaling.

Traditional chemotherapeutic agents commonly used for breast cancer treatment such as cyclophosphamide and paclitaxel (Taxol) often become ineffective due to chemo-resistance, particularly under hypoxic condition [25]. Recent investigations indicate the role of HIF1, a master transcriptional regulator for adaptation of tumor cells to hypoxia, in the development of drug resistance by aiding tumor cells escaping from drug-induced apoptosis or DNA damage-repair events [30–33]. It was also reported that Taxol treatment of tumor cells produced ROS which caused stabilization of HIF1 followed by induction of drug resistance [34]. In this report, we have shown that low dose of cacalol in combination with Taxol significantly blocked the generation of ROS and inhibited FAS expression, which in turn induced apoptosis in a synergistic manner in Taxol-resistant cell line, and that this combination of these compounds showed significantly more potent tumor suppressive activity than each compound alone in vivo. The results indicate that the Taxol-resistance is partially overcome with the treatment of cacalol, and therefore, supports the notion that FAS inhibition with a combination of other drugs serves as a rational strategy for the treatment of chemo-resistant tumors as we previously reported [15].

In summary, we identified cacalol as a novel and potent anti-tumor agent which blocks the FAS expression through modulation of Akt-SREBP pathway in human breast cancer cells. Because cacalol is derived from an edible plant and it showed evident anti-cancer effects in our animal experiments without apparent toxicity, cacalol is considered to be a promising candidate for chemopreventive as well as chemotherapeutic agent for breast cancer.

References

1. Kuhajda FP, Jenner K, Wood FD et al (1994) Fatty acid synthesis: a potential selective target for antineoplastic therapy. *Proc Natl Acad Sci USA* 91:6379–6383
2. Kuhajda FP (2000) Fatty-acid synthase and human cancer: new perspectives on its role in tumor biology. *Nutrition* 16:202–208
3. Menendez JA, Lupu R (2004) Fatty acid synthase-catalyzed de novo fatty acid biosynthesis: from anabolic-energy-storage pathway in normal tissues to jack-of-all-trades in cancer cells. *Arch Immunol Ther Exp (Warsz)* 52:414–426
4. Kuhajda FP, Katumuluwa AI, Pasternack GR (1989) Expression of haptoglobin related protein and its potential as a tumor antigen. *Proc Natl Acad Sci USA* 86:1188–1192
5. Kuhajda FP, Piantadosi S, Pasternack GP (1989) Haptoglobin-related protein (Hpr) epitopes in breast cancer as a predictor of recurrence of the disease. *N Engl J Med* 321:636–641
6. Van de Sande T, De Schrijver E, Heyns W, Verhoeven G, Swinnen JV (2002) Role of the phosphatidylinositol 3'-kinase/PTEN/Akt kinase pathway in the overexpression of fatty acid synthase in LNCaP prostate cancer cells. *Cancer Res* 62:642–646
7. Yang YA, Han WF, Morin PJ, Chrest FJ, Pizer ES (2002) Activation of fatty acid synthesis during neoplastic transformation: role of mitogen-activated protein kinase and phosphatidylinositol 3-kinase. *Exp Cell Res* 279:80–90
8. Menendez JA, Mehmi I, Atlas E, Colomer R, Lupu R (2004) Novel signaling molecules implicated in tumor-associated fatty acid synthase-dependent breast cancer cell proliferation and survival: role of exogenous dietary fatty acid, p53–21WAF1/CIP1, ERK1/2 MAPK, p27KIP1, BRCA1, and NF-kappaB. *Int J Oncol* 24:591–608
9. Porstmann T, Griffiths B, Chung YL et al (2005) PKB/Akt induces transcription of enzymes involved in cholesterol and fatty acid biosynthesis via activation of SREBP. *Oncogene* 24:6465–6481
10. Shimano H, Yahagi N, Amemiya-Kudo M et al (1999) Sterol regulatory element binding protein-1 as a key transcription factor for nutritional induction of lipogenic enzyme genes. *J Biol Chem* 274:35829–35832
11. Wang D, Sul HS (1998) Insulin stimulation of the fatty acid synthase promoter is mediated by the phosphatidylinositol 3-kinase pathway: involvement of protein kinase B/Akt. *J Biol Chem* 273:25420–25426
12. Fleischmann M, Iynedjian PB (2000) Regulation of sterol regulatory-element binding protein 1 gene expression in liver: role of insulin and protein kinase B/cAkt. *Biochem J* 349:13–17
13. Kotzka J, Muller-Wieland D, Roth G et al (2000) Sterol regulatory element binding proteins (SREBP)-1a and SREBP-2 are linked to the MAP-kinase cascade. *J Lipid Res* 41:99–108
14. Baron A, Migita T, Tang D, Loda M (2004) Fatty acid synthase: a metabolic oncogene in prostate cancer? *J Cell Biochem* 91:47–53
15. Furuta E, Pai S, Zhan R et al (2008) Fatty acid synthase gene is up-regulated by hypoxia via activation of Akt and sterol regulatory element binding protein-1. *Cancer Res* 68:1003–1011

16. Bandyopadhyay S, Zhan R, Wang Y et al (2006) Mechanism of apoptosis induced by the inhibition of fatty acid synthase in breast cancer cells. *Cancer Res* 66:5934–5940
17. Shindo K, Kimura M, Iga M (2004) Potent antioxidative activity of Cacalol, a sesquiterpene contained in *Cacalia delphiniifolia* Sleb et Zucc. *Biosci Biotechnol Biochem* 68:1393–1394
18. Kedrowski B, Hoppe R (2008) A concise synthesis of ($\pm$)-cacalol. *J Org Chem* 73:5177–5179
19. Fernandes-Alnemri T, Litwack G, Alnemri ES (1994) CPP32, a novel human apoptotic protein with homology to *Caenorhabditis elegans* cell death protein Ced-3 and mammalian interleukin-1 beta-converting enzyme. *J Biol Chem* 269:30761–30764
20. Naya K, Miyoshi Y, Mori H (1976) The sesquiterpenes of *Cacalia* species. *Chem Lett* 73–76
21. Torihata A, Hanai R, Gong X (2007) Chemical and genetic differentiation of *Ligularia tsang-chenensis* in Yunnan and Sichuan provinces of China. *Chem Biodivers* 4:500–507
22. Inman W, Luo J, Jolad DS, King SR, Cooper R (1999) Antihyperglycemic sesquiterpenes from *Psacalium decompositum*. *J Nat Prod* 62:1088–1092
23. Jimenez-Estrada M, Chilpa RR, Apan TR (2006) Anti-inflammatory activity of cacalol and cacalone sesquiterpenes isolated from *Psacalium decompositum*. *J Ethnopharmacol* 105:34–38
24. Medes G, Thomas A, Weinhouse S (1953) Metabolism of neoplastic tissue IV: a study of lipid synthesis in neoplastic tissue slices in vitro. *Cancer Res* 13:27–29
25. Swinnen J, Brusselmans K, Verhoeven G (2006) Increased lipogenesis in cancer cells: new players, novel targets. *Curr Opin Clin Nutr Metab Care* 9:358–365
26. Hockel M, Vaupel P (2001) Tumor hypoxia: definitions and current clinical, biologic, and molecular aspects. *J Natl Cancer Inst* 93:266–276
27. Knowles LM, Axelrod F, Browne CD, Smith JW (2004) A fatty acid synthase blockade induces tumor cell-cycle arrest by down-regulating Skp2. *J Biol Chem* 279:30540–30545
28. Pizer ES, Jackisch C, Wood FD, Pasternack GR, Davidson NE, Kuhajda FP (1996) Inhibition of fatty acid synthesis induces programmed cell death in human breast cancer cells. *Cancer Res* 56:2745–2747
29. Thupari JN, Pinn ML, Kuhajda FP (2001) Fatty acid synthase inhibition in human breast cancer cells leads to malonyl-CoA-induced inhibition of fatty acid oxidation and cytotoxicity. *Biochem Biophys Res Commun* 285:217–223
30. Unruh A, Ressel A, Mohamed HG et al (2003) The hypoxia-inducible factor-1 α is a negative factor for tumor therapy. *Oncogene* 22:3213–3220
31. Erler JT, Cawthorne CJ, Williams KJ et al (2004) Hypoxia-mediated downregulation of bcl-2 and bax in tumors occurs via hypoxia-inducible factor 1-dependent and -independent mechanisms and contributes to drug resistance. *Mol Cell Biol* 24:2875–2889
32. Brown LM, Cowen RL, Debray C et al (2006) Reversing hypoxic cell chemoresistance in vitro using genetic and small molecule approaches targeting hypoxia inducible factor-1. *Mol Pharmacol* 69:411–418
33. Hussein D, Estlin EJ, Dive C, Makin GWJ (2006) Chronic hypoxia promotes hypoxia-inducible factor-1 α -dependent resistance to etoposide and vincristine in neuroblastoma cells. *Mol Cancer Ther* 5:2241–2250
34. Kim HS, Oh JM, Jin DH, Yang KH, Moon EY (2008) Paclitaxel induces vascular endothelial growth factor expression through reactive oxygen species production. *Pharmacology* 81:317–324